INCH POUND

MIL-M-38510/122A
AMENDMENT 2
12 February 1997
SUPERSEDING
AMENDMENT 1
16 November 1994

MILITARY SPECIFICATION

MICROCIRCUITS, LINEAR, HIGH SLEW RATE OPERATIONAL AMPLIFIERS, MONOLITHIC SILICON

Inactive for new design after 13 July 1995

This amendment forms a part of MIL-M-38510/122A, dated 15 September 1989, and is approved for use by all Departments and Agencies of the Department of Defense

PAGE 5

4.4.1c, delete and substitute the following:

"c. Subgroup 12 shall be added to group A inspection as specified in table III herein. Subgroup 12 shall be performed for initial qualification and after major process or design changes using a sample of 15 devices with no failure allowed."

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TABLE I, Input bias current ($+I_{IB}$), T_A + +25° only, device types 07 and 08 only, delete and substitute the following:

Input bias current 2/ 4/	$+I_{IB}$	$+T_A$ = +25°C $+V_{CM}$ = -10 V, 0 V	07,08	-.2	+.2	nA
		$+T_A$ = +25°C $+V_{CM}$ = +10 V		-1.0	+1.0	

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TABLE I, Input bias current ($-I_{IB}$), T_A + +25° only, device types 07 and 08 only, delete and substitute the following:

Input bias current 2/ 4/	$-I_{IB}$	$+T_A$ = +25°C $+V_{CM}$ = -10 V, 0 V	07,08	-.2	+.2	nA
		$+T_A$ = +25°C $+V_{CM}$ = +10 V		-1.0	+1.0	

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* FIGURE 2, Output resistors R21 and R24, delete the values of 30 ohms and substitute 15 ohms.

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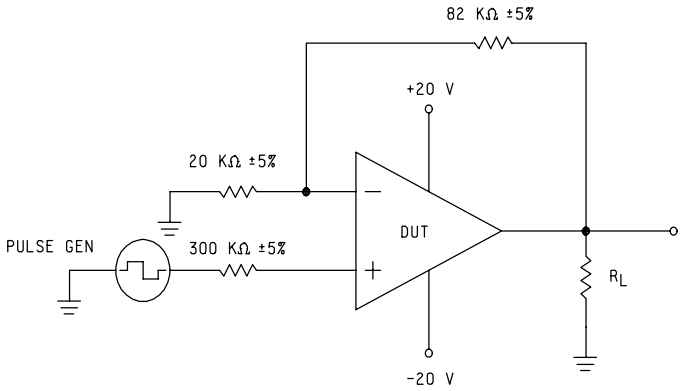
FIGURE 5, add the following figure:

"Device types 07 and 08



"

FIGURE 5, NOTE 1b, delete and substitute the following:

"b. ±V = 5.0 V for device types 01 through 06 and ±V = 1.0 V for device types 07 and 08."

FIGURE 5, NOTE 3, add the following sentence:

"R_L = 2 kΩ ±5% for device types 07 and 08."

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FIGURE 6, +PSRR, equation column, delete and substitute the following:

"+PSRR = 20 log 10 (10^4 / E13 - E14)".

FIGURE 6, -PSRR, equation column, delete and substitute the following:

"-PSRR = 20 log 10 (10^4 / E15 - E16)".

FIGURE 6, CMRR, equation column, delete and substitute the following:

"CMRR = 20 log 10 (2 x 10^4 / E17 - E18)".

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FIGURE 6, +PSRR, equation column, delete and substitute the following:

"+PSRR = 20 log 10 (10^4 / E13 - E14)".

FIGURE 6, -PSRR, equation column, delete and substitute the following:

"-PSRR = 20 log 10 (10^4 / E15 - E16)".

FIGURE 6, CMRR, equation column, delete and substitute the following:

"CMRR = 20 log 10 (2 x 10^4 / E17 - E18)".

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FIGURE 6, +PSRR, equation column, delete and substitute the following:

"+PSRR = 20 log 10 (10^4 / E13 - E14)".

FIGURE 6, -PSRR, equation column, delete and substitute the following:

"-PSRR = 20 log 10 (10^4 / E15 - E16)".

FIGURE 6, CMRR, equation column, delete and substitute the following:

"CMRR = 20 log 10 (2 x 10^4 / E17 - E18)".

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FIGURE 6, I_{IO}, equation column: Delete "I_{IO} = (E1-E4)/1000 /1 MΩ" and substitute "I_{IO} = (E4-E1)/1000 /10 MΩ";
delete "I_{IO} = (E2-E5)/1000 /1 MΩ" and substitute "I_{IO} = (E5-E2)/1000 /10 MΩ";
delete "I_{IO} = (E3-E6)/1000 /1 MΩ" and substitute "I_{IO} = (E6-E3)/1000 /10 MΩ";

FIGURE 6, $+I_{IB}$, equation column: Delete "$+I_{IB}$ = (E1-E7)/1000 /1 MΩ" and substitute "$+I_{IB}$ = (E7-E1)/1000 /10 MΩ";
delete "$+I_{IB}$ = (E2-E8)/1000 /1 MΩ" and substitute "$+I_{IB}$ = (E8-E2)/1000 /10 MΩ";
delete "$+I_{IB}$ = (E3-E9)/1000 /1 MΩ" and substitute "$+I_{IB}$ = (E9-E3)/1000 /10 MΩ";

FIGURE 6, $-I_{IB}$, value column: Delete "E10" "E11" and "E12".

FIGURE 6, $-I_{IB}$, units column: Delete "V" three times.

FIGURE 6, $-I_{IB}$, equation column: Delete "$-I_{IB}$ = (E10-E1)/1000 /1 MΩ" and substitute "$-I_{IB}$ = (E4+E7)/1000 /10 MΩ";
delete "$-I_{IB}$ = (E11-E2)/1000 /1 MΩ" and substitute "$-I_{IB}$ = (E5+E8)/1000 /10 MΩ";
delete "$-I_{IB}$ = (E12-E3)/1000 /1 MΩ" and substitute "$-I_{IB}$ = (E6+E9)/1000 /10 MΩ";

FIGURE 6, CMRR, equation column, delete and substitute the following: "CMRR = 20 log (20 / $\frac{E3 - E2}{1000}$)"

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FIGURE 6, $+A_{VS}$, equation column, delete and substitute the following; "$+A_{VS}$ = 10/(E20-E19/1000)"

FIGURE 6, $-A_{VS}$, equation column, delete and substitute the following; "$+A_{VS}$ = 10/(E21-E19/1000)"

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FIGURE 6, NOTE 1, delete and substitute the following:

"1. At T_A = +125°C, S7 and S8 shall be closed and the factor of 1000 shall be replaced with a factor of 100 in all equations for I_{IO}, $+I_{IB}$,and $-I_{IB}$, and the factor of 10 MΩ shall be replaced with a factor of 1 MΩ in all equations for I_{IO}, $+I_{IB}$, and $-I_{IB}$. Also the units shall be nA."

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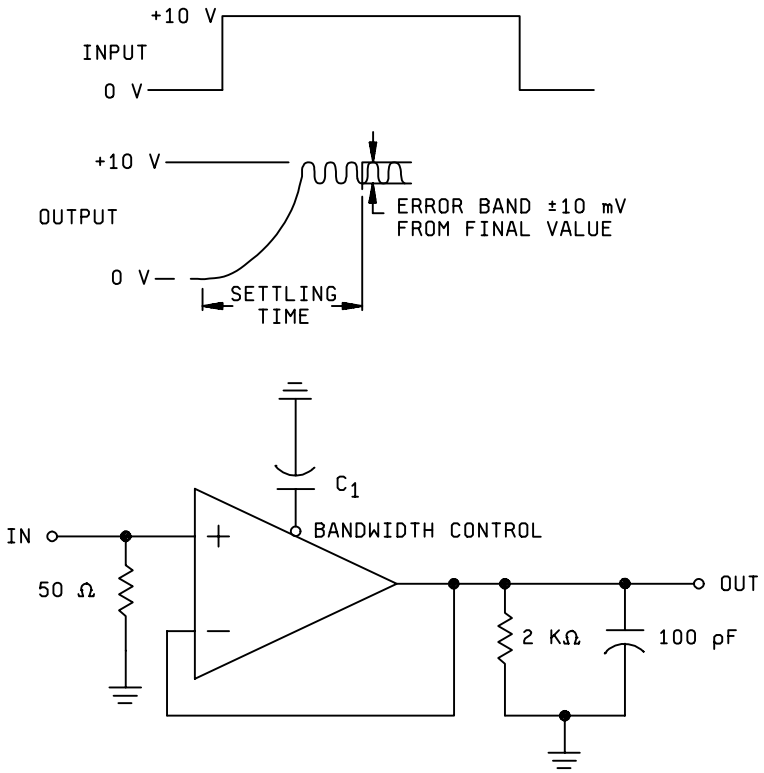
FIGURE 7, delete and substitute the following:

"Device type 02



NOTE: C1 is test fixture capacitance and shall be 1.5 pF ±1.0 pF.

FIGURE 7. Settling time circuits."

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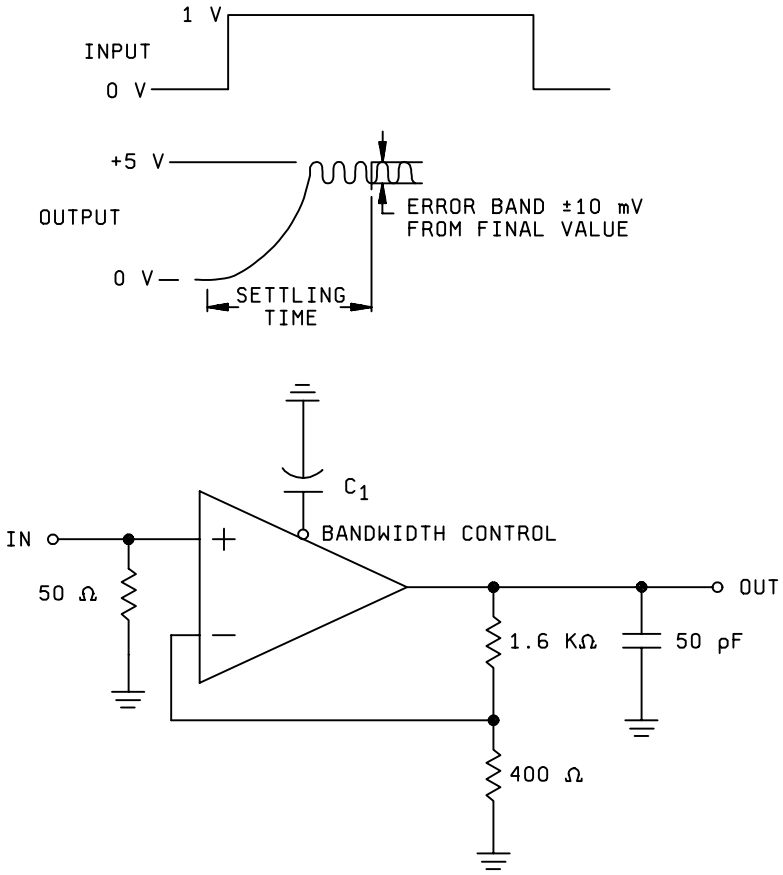
FIGURE 7, delete and substitute the following:

" Device type 03



NOTE: C1 is test fixture capacitance and shall be less than 1.0 pF.

FIGURE 7. Settling time circuits - Continued. "

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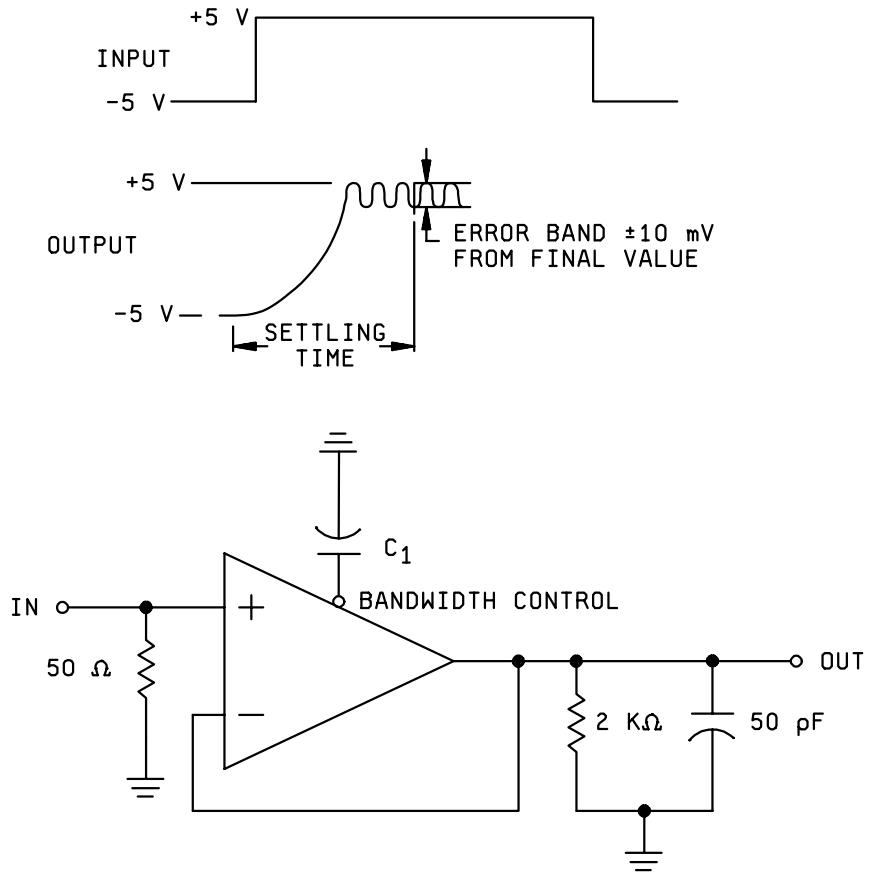
FIGURE 7, delete and substitute the following:

" Device types 04 and 05



NOTE: C1 is test fixture capacitance and shall be less than 1.0 pF.

FIGURE 7. Settling time circuits - Continued. "

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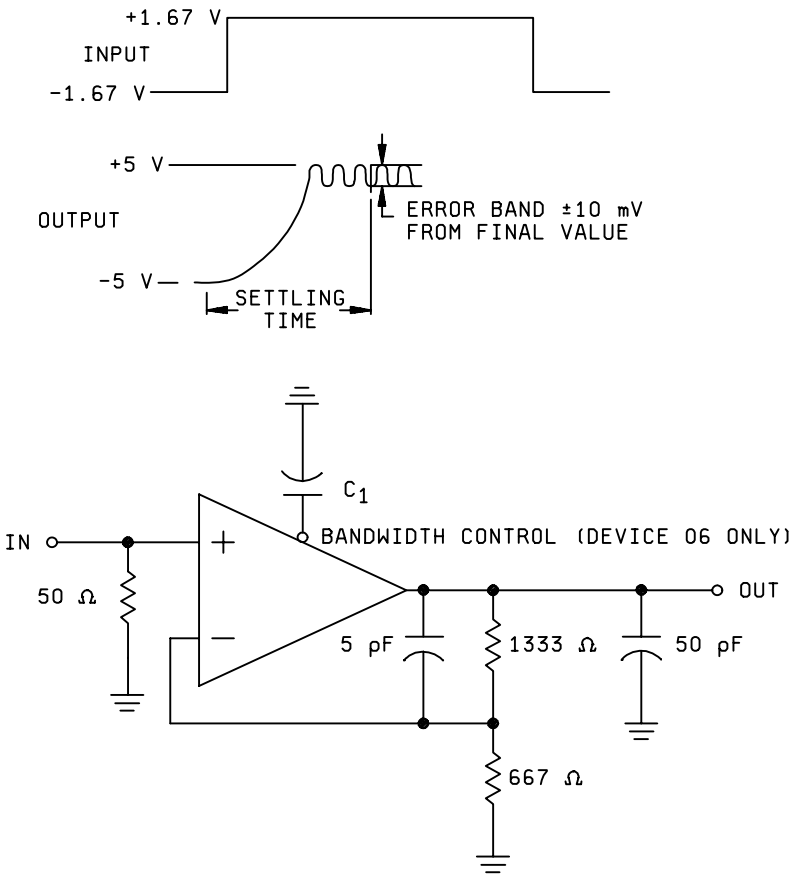
FIGURE 7, delete and substitute the following:

" Device types 06 and 07



NOTE: C1 is test fixture capacitance and shall be 2.5 pF ±1.0 pF.

FIGURE 7. Settling time circuits - Continued. "

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TABLE III, test numbers 9, 12, 27 and 30, device types 07 and 08 limits, minimum column:
Delete "-0.2" and substitute "-1.0".

TABLE III, test numbers 9, 12, 27 and 30, device types 07 and 08 limits, maximum column:
Delete "+0.2" and substitute "1.0".

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TABLE III, test numbers 45 and 48, device types 07 and 08 limits, minimum column:
Delete "-0.2" and substitute "-1.0".

TABLE III, test numbers 45 and 48, device types 07 and 08 limits, maximum column:
Delete "+0.2" and substitute "1.0".

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TABLE III, group A inspection, subgroup 12, test numbers 91 ($+t_S$) and 92 ($-t_S$), conditions column:
Delete "see figure 6" and substitute "see figure 7 14/".

TABLE III, group A inspection, subgroup 12, test numbers 91 ($+t_S$) and 92 ($-t_S$), device types 05, 06, 07 and 08 limits, maximum column: Delete "14/" and substitute "15/".

TABLE III, group A inspection, subgroup 13, test numbers 93 ($\Delta V_{IO}/\Delta T$), and 94 ($\Delta I_{IO}/\Delta T$), 95 ($\Delta V_{IO}/\Delta T$), and 96 ($\Delta I_{IO}/\Delta T$), conditions column: Delete "15/" and substitute "16/".

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TABLE III, footnotes: Delete "14/" and "15/" in their entirety.

TABLE III, after footnote 13/ add the following footnotes:

"14/ For device types 02, 03, 04, 05 and 06, which have a bandwidth control pin, no additional external compensation capacitance shall be used.

15/ At +25°C, tests 91 and 92, $+t_S$ and $-t_S$, device type 05: 01 μs maximum, device type 06, 07, and 08: 1.1 μs maximum.

16/ Test 93 through 96, which require a read and record measurement plus a calculation, may be omitted except when subgroup 13 is being accomplished for group A sampling inspection and group B (class S) inspection."

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TABLE IV, footnote 1/, add the following sentence: "For device types 07 and 08, delta limits apply to the measured value of test 1, 7 and 10."

TABLE IV, after footnote 2/, add footnote 3/ as follows:

"3/ For device types 07 and 08, limits apply for tests 1, 2, 3, 7, 8, 9, 10, 11, and 12."

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17

Review activities:
Army - AR, MI, SM
Navy - AS, CG, MC, OS, SH
Air Force - 19, 85, 99

Civil Agency Coordinating Activity:
NASA - NA

Preparing activity:
DLA - CC

(Project 5962-1733)

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